ARASH JALIL KHABBAZI

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SUMMARY

My research is on modeling, control, optimization, and machine learning for building energy systems. It advances the state of the art of practical automation systems for efficient, occupant- and grid-interactive buildings, focusing on reinforcement learning and data-driven predictive control. I am developing and demonstrating scalable algorithms in real buildings, minimizing the data and engineering needed to achieve good, if imperfect, performance. I also work with Purdue's Energy and Utilities team on demand flexibility in commercial buildings, and I develop power control systems for homes.

EDUCATION

Purdue University, *West Lafayette, IN* — *08/2023 – Present*
PhD in Mechanical Engineering — **GPA:** 3.9/4.0
Concentration in Computational Science and Engineering

University of British Columbia (UBC), *BC, Canada* — *05/2021 – 08/2023*
MS in Mechanical Engineering — **GPA:** 94% (4.0/4.0)
Thesis: "Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines"

University of Tabriz, *EA, Iran* — *08/2016 – 07/2020*
BS in Mechanical Engineering — **GPA:** 19.12/20 (4.0/4.0) – **Rank:** 1/155+

SELECT AWARDS AND RECOGNITION

ASHRAE Graduate Student Grant-in-Aid Award *2025*
Best PhD Forum Presentation Runner-up Award, *ACM BuildSys* *2025*
ACM SIGEnergy Student Travel Grant, *ACM BuildSys* *2025*
Best Paper Award, *CSME International Congress* *2023*
Best Presentation Award, *CSME International Congress (Advanced Energy Symposium)* *2022*
Graduate Research Scholarship, *University of British Columbia* *2022*
Graduate Dean's Entrance Scholarship, *University of British Columbia* *2021*

PUBLICATIONS

Journal Articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review." *Applied Energy, 2025.*
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions." *International Journal of Hydrogen Energy, 2024.*

Preprints

1. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher, "Indirect Data-Driven Predictive Control and the State-Space Predictor." *arXiv:2602.10936, 2026. Under review.*

Conference Papers

6. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher, "Scalable Supervisory HVAC Control for Linear Objectives." *65th IEEE Conference on Decision and Control (CDC), 2026.*
5. L.D. Reyes Premer, E.N. Pergantis, **A.J. Khabbazi**, F. Wu, and K.J. Kircher, "Evaluating Power Control Strategies for Avoiding Residential Panel Upgrades in Cold Climates." *ACEEE Summer Study on Energy Efficiency in Buildings, 2026.*
4. **A.J. Khabbazi** and K.J. Kircher, "Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings." *American Control Conference (ACC), 2026.*

3. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, "Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study." *CISBAT, 2025.*
2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, and K.J. Kircher, "What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?" *8th International High Performance Buildings Conference, 2024.*
1. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, "Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions." *Canadian Society for Mechanical Engineering (CSME) International Congress, 2023.* ***Best Paper Award.***

Abstracts

3. **A.J. Khabbazi**, "PhD Forum Abstract: Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment." *12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2025.* ***Best PhD Forum Presentation Runner-up Award.***
2. **A.J. Khabbazi**, R. Li, and J. Quinn, "Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid." *Canadian Society for Mechanical Engineering (CSME) International Congress, 2022.* ***Best Presentation Award.***
1. **A.J. Khabbazi**, R. Li, J. Quinn, and M. Hoorfar, "The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines." *13th International Green Energy Conference (IGEC-XIII), 2021.*

TALKS

10. "Advanced HVAC Control in Real Buildings: State of the Art, Protocols, and Deployment." *Google Smart Buildings Simulator (SBSim) Group, 2026.*
9. "Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings." *American Control Conference (ACC), New Orleans, LA, 2026.*
8. "Model Predictive Control for Buildings." *Google Smart Buildings Simulator (SBSim) Group, 2026.*
7. "Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment." *PhD Forum, 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), Golden, CO, 2025.*
6. "What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?" *8th International High Performance Buildings Conference, Purdue University, West Lafayette, IN, 2024.*
5. "Field Demonstrations of Advanced Commercial HVAC Control." *Intelligent Building Operations Workshop, Purdue University, West Lafayette, IN, 2024.*
4. "Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines." *Herrick Graduate Seminar, Purdue University, West Lafayette, IN, 2023.*
3. "Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions." *CSME International Congress, Université de Sherbrooke, QC, Canada, 2023.*
2. "Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid." *CSME International Congress, University of Alberta, AB, Canada, 2022.*
1. "The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines." *13th International Green Energy Conference (IGEC-XIII), 2021.*

RESEARCH EXPERIENCE

Purdue University, Graduate Research Assistant, Kircher Research Group, *West Lafayette, IN* *08/2023–Present*

- Identified a grey-box thermal model of a 13-zone campus building from BAS data for predictive pre-cooling control.
- Developed a two-parameter, tuning-free HVAC controller with **up to 87% of optimal savings** in simulation (CDC'26).
- Built data-driven predictive control with the **lowest prediction error and failure rate** of all predictors (under review).
- Derived and tested a multi-zone model showing small HVAC demos **overstate savings** (ACC'26, under review).
- Developed current-limiting control for home appliances to **avoid $2k–$10k panel upgrades** (CISBAT'25, ACEEE'26).
- Led a review of **100+** field demonstrations of MPC and RL for HVAC (Applied Energy).

UBC, Graduate Research Assistant, TMMF Lab & FortisBC Renewable Gas, *BC, Canada* *05/2021–08/2023*

- Ran CFD simulations of hydrogen blending into natural gas pipelines; built ideal- and real-gas mixture models in C.

– Delivered **1** journal paper, **3** conference papers, an MS thesis, and **2** awards on a project that seeded the UBC H_2Lab.

INDUSTRY EXPERIENCE

Tesla, Inc., Applied ML Intern, *Sparks, NV* *05/2025–08/2025*

– Built MySQL pipelines, engineered features, and trained ML models (XGBoost, random forests, neural nets) to $\mathbf{R^2}$ **0.85**.
– Helped deploy a thermal imaging system for drive-unit inspection, replacing destructive laboratory testing.
– Delivered ML-driven quality gains projected to cut testing costs by over **$1M/year**; offered a **three-month extension**.

Purdue Energy & Utilities, Energy Systems Researcher, *West Lafayette, IN* *02/2026–Present*

– Designed and ran a **three-day pre-cooling trial** in a 13-zone campus building to shift cooling demand off-peak.
– Delivered a nine-year energy-use assessment of 21 renovations to **Purdue leadership** for the Giant Leaps Master Plan.
– Quantified ∼**$41k/yr** savings and a statistically significant **8%** lower approach from a two-plant air/dirt-separator retrofit.

TEACHING EXPERIENCE

Purdue University, Graduate Teaching Assistant (PhD)

– ME597: Distributed Energy Resources (DERs) *Spring'26*
  – Supported 52 students on modeling and control of DERs such as batteries, solar PV, EVs, and HVAC.
  – Assisted with homework, coding modules, and office hours covering DER concepts.
  – Guided semester projects on DER topics, including building controls and data center cooling.

University of British Columbia, Graduate Teaching Assistant (MS)

– Engineering Analysis I *Fall'21, Spring'22, Fall'22*
– Fluid Mechanics II Lab *Spring'22, Spring'23*

LEADERSHIP & MENTORSHIP

Research Mentor, Purdue University *2024–Present*

– Mentored three master's students and one undergraduate on research planning and technical writing.
– Collaborated on their research and co-authored the resulting publications.

Disability Resource Centre Invigilator, University of British Columbia *2021–2023*

– Invigilated accommodated exams, supporting students with approved testing accommodations.

ACADEMIC SERVICE

Conference and workshop organizing:

July 2026	Student Chair, Intelligent Building Operations (IBO) Workshop, *West Lafayette, IN*
July 2024	Student Chair, Intelligent Building Operations (IBO) Workshop, *West Lafayette, IN*

Society memberships: ASHRAE, ACM, IEEE, CSME

Reviewing: IEEE Transactions on Control Systems Technology; IEEE Conference on Decision and Control; Building and Environment; Journal of Building Performance Simulation; ACEEE Summer Study on Energy Efficiency in Buildings; Herrick Conferences

SELECT COURSES

- ECE570: Artificial Intelligence
- IE690: Reinforcement Learning & Control (RLC)
- AAE568: Applied Optimal Control & Estimation
- AAE 561: Intro to Convex Optimization
- STAT511: Statistical Methods
- ME597: Industrial Internet of Things (IIoT)
- ME597: Distributed Energy Resources (DERs)
- ME511: Analysis of Thermal Systems

SKILLS

Languages:	Python, MATLAB, C, SQL
Optimization & ML:	CVXPY/CVX, TensorFlow, scikit-learn, XGBoost, Stable-Baselines3, PyTorch, Gurobi
Simulation & Testbeds:	BOPTEST, Gymnasium, OCHRE
Systems & Tools:	Git, InfluxDB, Grafana, MySQL, Pandas, NumPy, Linux, Docker, LaTeX